# Abdul Raheem Shahzad

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## Education

**Wuhan University (QS Rank 165)** — Wuhan, China
M.S., Software Engineering. GPA: 3.67/4.0 (Ranked 2nd in class) — Sep 2024 – Jun 2026
Thesis: *Development and Evaluation of AI Models for Predicting IVF/ICSI Treatment Outcomes*
Honors: China Scholarship Council (CSC) Scholarship; Outstanding International Student Award

**CECOS University of IT and Emerging Sciences** — Peshawar, Pakistan
B.S., Software Engineering. CGPA: 3.95/4.0 (Ranked 1st in department, Gold Medalist) — Oct 2018 – Nov 2022
Thesis: *Machine Learning Based Leukemia Cancer Prediction Using Protein Sequential Data*

## Research & Professional Experience

**Zhongnan Hospital of Wuhan University** — Wuhan, China
**Clinical Research Collaborator** — Sep 2025 – Present

- Collaborate with Prof. Zhang's clinical team on machine-learning prediction of IVF/ICSI live-birth outcomes.
- Build the models and evaluation pipeline that form the basis of the M.S. thesis and two first-author manuscripts.

**Self-Employed (Freelance)** — Remote
**Data Science Instructor / Academic Research Support Specialist** — Feb 2023 – Sep 2024

- Taught data analysis, machine learning, and deep learning to a diverse group of students.
- Wrote research code to produce results aligned with students' thesis methodologies.

**Codanics** — Faisalabad, Pakistan (Remote)
**Data Science Instructor** — Sep 2022 – Feb 2023

- Instructed new students in data science and machine learning (lecture series).
- Deployed ML-powered dashboards and advised researchers on applying AI in their projects.

## Publications

[1] **A. R. Shahzad**†, J. Chen†, M. Zhang, J. Deng. "Predicting Live Birth After Fresh Embryo Transfer in IVF/ICSI: Differential ICM/Trophectoderm Morphology Encoding and a Hybrid Attention–Logistic Regression Ensemble." *IEEE Journal of Biomedical and Health Informatics (JBHI)*, 2026. Preprint: arXiv:2610.21134.

[2] **A. R. Shahzad**, M. Z. Khan, U. Ajaib, A. U. Rehman, J. Deng, M. Zhang. "Machine Learning for Live-Birth Prediction After IVF/ICSI From Structured Clinical Data: A Systematic Review, Meta-Analysis, and Critical Appraisal." Under review, *IEEE TPAMI*, 2026. Preprint: arXiv:2611.07124.

[3] M. Z. Khan, X. Li, D. Ali, Y. Xu, **A. R. Shahzad**. "Detecting Unsupported Claims in Audience-Adaptive Clinical Trial Summaries: A Claim-Level Benchmark." Preprint, 2026. OpenReview.

[4] S. Ahmad, L. Song, K. Wanghe, Z. Wu, G. Chen, I. Strelnikov, L. Bosso, **A. R. Shahzad**, W. Mao. "Climate Change Impacts Indo-Pacific Seagrass Beds: Insights from Ecological Niche Modelling and Phylogenetic Analyses." *Ecological Indicators* (Elsevier), in press, 2026.

[5] W. Ahmad, **A. R. Shahzad**, M. A. Amin, W. H. Bangyal, T. J. Alahmadi, S. H. Khan. "Machine Learning Driven Dashboard for Chronic Myeloid Leukemia Prediction Using Protein Sequences." *PLoS One*, 20(6):e0321761, 2025. DOI.

[6] R. Dars, J. Ping, X. Mei, C. Chen, **A. R. Shahzad**, J. Mahwa, M. A. Jamali. "Predicting Climate Change Impacts on Groundwater Aquifer Levels in the Henan North China Plain." *Applied Water Science*, 15(6):120, 2025. DOI.

†Equal contribution.

## Honors & Awards

**Outstanding International Student Award**, School of Computer Science, Wuhan University — Dec 2025
**China Scholarship Council (CSC) Scholarship**, fully funded M.S. study in China — Jul 2024
**Winner, Open Doors Olympiad**, Data Science & AI track (list, certificate) — Feb 2023
**Gold Medal**, highest grades in the department, CECOS University (certificate) — Dec 2022
**Silver Medal**, Kaggle Data Science & AI competition (notebook) — Nov 2022
**Shield of Honor**, Resource Person, Data Science with Python in Agriculture workshop, UAF (certificate) — Nov 2022

**Best Computer Science Project**, final-year project (later published in a Q1 journal) (certificate) Sep 2022
**Merit-Based Scholarship**, CECOS University, awarded in all but one semester 2018 – 2022
**Kaggle Dataset & Notebook Expert** (profile)

## Selected Projects

- **Leukemia Prediction from Protein Sequences** (B.S. final-year project): built a Streamlit dashboard predicting leukemia from protein sequential data; XGBoost reached 94.5% accuracy. [GitHub]
- **MONAI MedNIST Classification**: trained a 6-class medical image classifier (CT, MRI, X-ray, etc.) in PyTorch/MONAI with mixed precision, gradient accumulation, early stopping, and LR scheduling. [GitHub]
- **LLM (DistilBERT) vs. GAN-Synthetic Data**: compared DistilBERT trained on original data against the same model trained on GAN-generated synthetic data. [Link]
- **Nuclei Image Segmentation with U-Net**: segmented nuclei across varied cell types, magnifications, and imaging modalities (brightfield vs. fluorescence). [GitHub]
- **YBVPD – YOLOv7 Visual Pollution Detection**: detected visual pollution across Saudi Arabian cities for the government-hosted Smartathon. [GitHub]
- **Generative Adversarial Networks**: implemented DCGAN, WGAN, WGAN-GP, and Conditional GAN on MNIST. [GitHub]
- **Transfer Learning on FoodVision**: fine-tuned EfficientNet in PyTorch. [GitHub]
- **Nitrate Leaching Prediction in Irrigated Agriculture**: built a prediction dashboard from research data. [GitHub]
- **Data Augmentation for Breast Cancer Prediction**: analyzed the impact of augmentation on tabular classification. [Kaggle]
- **MBTI Personality Prediction App**: predicted MBTI types and deployed a Streamlit web app. [GitHub]
- **Airline Time-Series Hackathon (Sastaticket.pk)**: performed EDA on arrival/departure data to find optimal booking windows. [GitHub]
- **Meta-Estimators with scikit-learn** (Jan 2023). [GitHub]

## Conferences, Workshops & Talks

**ICMSS 2026** Wuhan, China
Participant, 10th Intl. Conf. on Management Engineering, Software Engineering & Service Sciences (certificate) Mar 2026

**"Data Delights: Pie & AI Interactive Analytics" – DeepLearning.AI** Online
Workshop Instructor, invited by a DeepLearning.AI Ambassador (post) Jul 2023

**Data Science with Python in Agriculture, University of Agriculture Faisalabad** Faisalabad, Pakistan
Resource Person; taught machine learning to faculty and students in a three-day workshop (certificate) Nov 2022

## Leadership & Activities

**Wuhan University** Wuhan, China
**Country Representative (Pakistan)** Dec 2025 – Present

- Represent Pakistani culture, cuisine, and traditions at university international events.

**CECOS Blood Society, CECOS University** Peshawar, Pakistan
**Team Member** Mar 2020 – Aug 2024

- Arranged blood donors for emergency and routine cases.

## Skills

**Programming:** Python, MATLAB, C/C++, R, PHP, HTML/CSS
**ML / AI:** PyTorch, TensorFlow, Machine Learning, Deep Learning, Computer Vision, NLP, LLM Fine-Tuning, Generative AI, Agentic AI
**Tools:** Git/GitHub, VS Code, Cursor, TRAE, Google Colab, TensorBoard, MS Office

## References

**Prof. Kifayat Ullah**, Head of Department, Department of Computer Science, CECOS University • kifayat@cecos.edu.pk
**Dr. Zhang Yun**, Associate Professor, Department of Computer Science, Wuhan University • yunzhang@whu.edu.cn